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Hallucination Detection in LLMs via Beam Search Sampling and Semantic Consistency Analysis

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***Abstract*—Large Language Models (LLMs) have revolutionized natural language processing, enabling high-quality content generation. However, they remain prone to hallucinations, instances where generated outputs deviate from factual truth. In this paper, we propose a novel approach that integrates Beam Search Sampling (`BSS`) with Semantic Consistency Analysis to systematically detect factual hallucinations. Our method leverages `BSS` to generate multiple candidate answers, capturing the model's confidence distribution across different plausible answers. These answers are then clustered based on semantic similarity, and a Natural Language Inference ($\mathcal{NLI}$) model is applied to assess entailment and contradiction relationships. To quantify hallucinations, we introduce a scoring mechanism that combines token probabilities with semantic similarity metrics. Additionally, for cases where `BSS` produces a single answer, we incorporate a Chain-of-Verification (`CoVe`) mechanism to perform self-consistency checks. We evaluate our approach using Llama3.8B on the TruthfulQA dataset, achieving a precision of 0.87 and an AUROC of 0.81 for multi-answer generation. For single-answer verification with CoVe, we report a precision of 0.64 and accuracy of 0.71. Our approach outperforms conventional semantic entropy-based methods.**

***Index Terms*—Large Language Models (LLMs), Hallucination Detection, Beam Search Sampling, Semantic Consistency.**

I. Introduction

Large Language Models (LLMs), such as `GPT-4` [21], `Gemini` [3], and `Llama` [7], have transformed natural language processing by combining various tasks as a unified sequence generation problem. This approach allows users to engage with a wide range of applications, including question answering, code generation, and mathematical problem solving, through an intuitive natural language interface. These models offer not only remarkable performance, but also enhanced interactivity. In addition, their extensive factual knowledge has significantly improved information retrieval capabilities [1], potentially redefining how information search systems operate [25].

While the generative paradigm of LLMs offers substantial benefits, enabling efficient and versatile content generation across various domains, it also introduces certain challenges. The generated content may sometimes contain inaccuracies or deviate from factual truth due to the inherent limitations of LLMs [11]. This phenomenon, referred to as factual hallucinations, occurs when generated content strays from verifiable real-world facts [9]. This problem presents a significant issue in high-stakes areas such as healthcare [4], finance [20], and law. For instance, in February 2025, Australian courts faced significant issues when lawyers used LLM to prepare legal documents, resulting in the inclusion of 17 non-existent case citations in an immigration appeal submission [8]. Such instances underscore the urgent need for robust methods to detect and mitigate factual hallucinations.

Current approaches for detecting factuality hallucinations can be categorized into two main categories. *External-retrieval fact checking*, which rely on retrieving information from external sources to validate model outputs [2, 10, 18]. These approaches are constrained by the quality and breadth of the external data available. In contrast, *zero-resource methods* do not rely on external data [5, 12, 15, 22]. One such approach is uncertainty estimation [13, 16], which utilizes semantic entropy to quantify uncertainty within the semantic space of model-generated responses. This is typically achieved by employing stochastic response generation techniques, such as adjusting the temperature of the LLM to produce diverse outputs. Although this method facilitates the generation of multiple answers, it does not provide a reliable measure of the LLM's confidence, as the generated responses are still subject to inherent variability. Moreover, these techniques can significantly increase latency in the model's response time, as generating multiple answers from scratch requires additional processing.

In this paper, we propose a novel approach to detecting factual hallucinations in LLM-generated answers by integrating Beam Search Sampling (`BSS`) with Semantic Consistency Analysis. The key steps are illustrated in Fig. 1. The `BSS` generates multiple candidate answers when the model exhibits uncertainty, where new beams (answers) are created when the model exhibits uncertainty in its generated tokens. These answers are then clustered based on semantic similarity, and an $\mathcal{NLI}$ model is used to evaluate entailment and contradiction relationships. Additionally, when `BSS` produces a single answer, we incorporate a `CoVe` self-checking mechanism to assess the consistency of the answer by generating verification questions and evaluating their alignment with the original output. We evaluated our approach using `Llama 3.8B` as the underlying LLM and `TruthfulQA` as the benchmark dataset. The results demonstrate that our method achieves an AUROC of **0.81** when `BSS` generate multiple answers. For `CoVe` self-checking, we obtain a precision of **0.64** and an accuracy of **0.71**.

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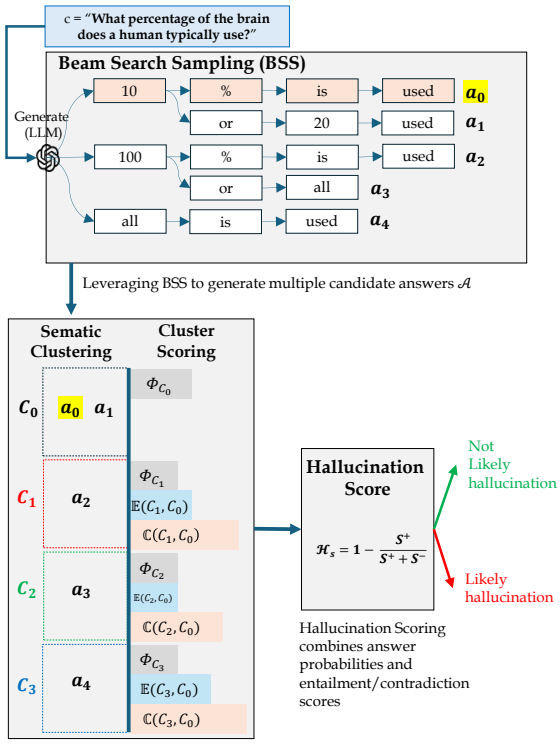


Fig. 1: An overview of the hallucination detection approach. It starts by generating multiple candidate answers using `BSS`, where new beams are created when the model exhibits uncertainty in its generated tokens. The generated answers are then clustered based on semantic similarity. To assess consistency, an $\mathcal{NLI}$ model computes entailment $\mathbb{E}(C_i, C_0)$ and contradiction $\mathbb{C}(C_i, C_0)$ scores between clusters and the LLM-generated answer (a_0). The hallucination score $\mathcal{H}_S$ is then determined by combining the cluster probability score Φ_{C_i} with entailment and contradiction scores, identifying potential factual hallucinations.

In summary, our contributions are: (i) Beam Search Sampling for effective LLM-based answer generation, (ii) a hallucination score that integrates answer probability and cluster entailment/contradiction scores, and (iii) an evaluation of the proposed approach using the `Llama-3.8B` model on the `TruthfulQA` dataset.

II. Related Work

Current approaches for factuality hallucination detection involves assessing whether the output of LLMs aligns with real-world facts. Typical methods generally fall into two main categories:

External-retrieval fact checking: one of the most straightforward approaches to fact verification involves retrieving information from external sources. In [18] authors introduced FACTSCORE, a metric designed for assessing the factual accuracy of long-form text. Their method breaks down generated content into discrete factual units and evaluates the proportion supported by trusted knowledge sources. Building on this idea, Chern et al. [2] developed a unified framework that enhances LLMs' ability to detect factual inconsistencies by leveraging a suite of external tools for evidence retrieval. Huo et al. [10] further refined this process by incorporating query expansion, where the original question is combined with the LLM-generated response. This technique mitigates topic drift and improves the alignment between retrieved evidence, the initial query, and the model's output. However, these methods are inherently limited by their reliance on the quality and scope of available external data, which may not always capture every nuance or context, potentially leading to incomplete or biased evaluations.

Zero-resource methods: this can further be divided into: (i) *Internal checking:* this approach rests on the premise that LLMs encode a vast repository of factual information within their parameters. As a result, researchers are investigating how to harness this intrinsic knowledge to perform fact-checking, potentially streamlining verification processes without relying solely on external data sources. For example, Dhuliawala et al. [5] proposed the Chain-of-Verification (CoVe) approach, where an LLM initially generates a set of verification questions based on a draft answer and then uses its internal knowledge to check the consistency of the response, thus highlighting potential inaccuracies. Similarly, Kadavath et al. [12] and Zhang et al. [23] assess the factual correctness for the Boolean questions by computing the probability $\mathbb{P}(True)$ entirely from the model's internal data. Furthermore, Li et al. [14] observed that many individual factual statements are interconnected, with some providing essential context for others, which can lead to misjudgments if evaluated in isolation. To overcome this, they direct the LLM to evaluate hallucinations by taking into account all related factual statements. However, because LLMs are not designed to serve as definitive factual databases [24], relying solely on their inherent knowledge for fact checking can sometimes result in inaccurate conclusions. (ii) *Uncertainty estimation:* alternative strategies tap into the intrinsic processes of LLMs to extract meaningful indicators of their confidence, often reflected through measures such as token probability and entropy. For example, Varshney et al. [22] quantified uncertainty by analyzing the minimum token probability associated with key concepts, arguing that a low token probability robustly indicates the model's uncertainty regardless of other higher probability tokens. In a similar vein, Luo et al. [15] introduced a self-assessment approach for estimating uncertainty, suggesting that a model's ability to accurately reconstruct an original concept from its explanation serves as an indicator of its proficiency with that concept. Our proposed approach is similar to the one presented by Manakul et al. [16] and Kossen et al. [13], who detect hallucinations by comparing multiple responses generated from the same prompt to evaluate the consistency of factual statements. Both approaches rely on the quality of the responses generated. In [13], the authors use prompting techniques and adjust the LLM's temperature to produce diverse answers. While this approach facilitates the generation of multiple responses, it does not guarantee the model's confidence, as the answers remain subject to variability introduced by the temperature setting.

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III. Methodology

Our proposed approach systematically assesses factual hallucination in LLM-generated responses. As detailed in Algorithm 1, it takes as input an LLM $\mathcal{M}$, prompt tokens c, beam size b, and an $\mathcal{NLI}$ model. The *Hallucination Detection* algorithm then outputs a hallucination score $\mathcal{H}_\mathcal{S}$. First, the algorithm invokes BSS (details of the algorithm in Algorithm 2) to generate multiple candidate answers $\mathcal{A}$ (Line 4 of Algorithm 1). Each answer $a_i \in \mathcal{A}$ is assigned a probability score using the normalized geometric mean of token probabilities (Lines 5–7, Equation 1). The geometric mean mitigates sequence length bias by averaging log probabilities before exponentiation, ensuring that low-probability tokens significantly impact the final answer probability [19].

$$\mathbb{P}(a_i \mid c) = \exp\left(\frac{1}{|a_i|}\sum_{j=1}^{|a_i|}\log \mathbb{P}(t_j \mid t_{0:j-1}, c)\right) \quad (1)$$

Next, the algorithm selects the most probable answers (Line 8 of Algorithm 1) and clusters them based on semantic similarity (Line 9 of Algorithm 1, details of the algorithm in Algorithm3). For each cluster $\mathcal{C}_i$, it computes the cluster probability score $\Phi_{\mathcal{C}_i}$ and evaluates the entailment $\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c)$ and contradiction $\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c)$ scores using an $\mathcal{NLI}$ model (Lines 10–12 of Algorithm 1, details of the algorithm in Algorithm 4). If multiple clusters are identified (Line 13-26 of Algorithm 1), the hallucination score $\mathcal{H}_\mathcal{S}$ is then computed as follows:

$$\mathcal{H}_\mathcal{S} = 1 - \frac{S^+}{S^+ + S^-} \quad (2)$$

where S^+ and S^- represent the aggregated scores of clusters supporting and contradicting the LLM's response, respectively. Indeed, to determine whether a cluster supports or contradicts the LLM's response, we apply thresholds to the NLI model's outputs: a cluster is considered entailing if the entailment score exceeds $\tau_e = 0.5$, and contradicting if the contradiction score exceeds $\tau_c = 0.5$. These threshold values were empirically chosen. If only a single cluster is formed, the algorithm invokes the $SelfCheck$ module (Line 28) to further assess the factual consistency of the generated answer (see Section III-D).

A. Beam Search Sampling

Existing black-box hallucination detection methods typically generate multiple responses from LLMs by adjusting decoding parameters (e.g., modifying temperature) or designing prompts to encourage output variation [6, 13, 16]. However, these approaches often produce limited response diversity, restricting the ability to accurately assess the model's confidence. To address this limitation, we introduce BSS, a method that generates multiple candidate answers while capturing the model's confidence distribution across different plausible responses (at token level). The key steps of BSS are outlined in Algorithm 2.

Algorithm 1 *Hallucination Detection*

1: **Input:** LLM $\mathcal{M}$, $c = (c_0, c_1, \ldots, c_k)$, Beam size b, $\mathcal{NLI}$ model
2: **Constants:** τ_e (entailment threshold), τ_c (contradiction threshold)
3: **Output:** Hallucination Score $\mathcal{H}_\mathcal{S}$
4: $\mathcal{A} \leftarrow \texttt{BSS}(\mathcal{M}, c, b, \mathcal{NLI})$
5: **for** $a_i \in \mathcal{A}$ **do**
6: $\quad \mathbb{P}(a_i \mid c) \leftarrow \exp\left(\frac{1}{|a_i|}\sum_{j=1}^{|a_i|}\log \mathbb{P}(t_j \mid t_{0:j-1}, c)\right)$
7: **end for**
8: $\mathcal{A} \leftarrow \text{Top}_a(\mathcal{A})$
9: $\mathcal{C} \leftarrow SemanticClustering(\mathcal{A})$
10: **for** $\mathcal{C}_i \in \mathcal{C}$ **do**
11: $\quad \Phi_{\mathcal{C}_i}, \mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c), \mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c) \leftarrow ClusterScore(\mathcal{C}_0, \mathcal{C}_i, \mathcal{NLI})$
12: **end for**
13: **if** $|\mathcal{C}| > 1$ **then**
14: $\quad S^+ \leftarrow \Phi_{\mathcal{C}_i} \cdot \mathbb{E}(\mathcal{C}_0 \mid \mathcal{C}_0, c)$
15: $\quad S^- \leftarrow \Phi_{\mathcal{C}_i} \cdot \mathbb{C}(\mathcal{C}_0 \mid \mathcal{C}_0, c)$
16: $\quad$ **for** $i = 1$ **to** $|\mathcal{C}|$ **do**
17: $\quad\quad$ **if** $\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c) > \tau_e$ **then**
18: $\quad\quad\quad S^+ \leftarrow S^+ + \Phi_{\mathcal{C}_i} \cdot \mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c)$
19: $\quad\quad$ **else if** $\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c) > \tau_c$ **then**
20: $\quad\quad\quad S^- \leftarrow S^- + \Phi_{\mathcal{C}_i} \cdot \mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c)$
21: $\quad\quad$ **else**
22: $\quad\quad\quad S^+ \leftarrow S^+ + \Phi_{\mathcal{C}_i} \cdot \mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c)$
23: $\quad\quad\quad S^- \leftarrow S^- + \Phi_{\mathcal{C}_i} \cdot \mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c)$
24: $\quad\quad$ **end if**
25: $\quad$ **end for**
26: $\quad \mathcal{H}_\mathcal{S} \leftarrow 1 - \frac{S^+}{S^+ + S^-}$
27: **else**
28: $\quad \mathcal{H}_\mathcal{S} \leftarrow SelfCheck(\mathcal{M}, c, \mathcal{A})$
29: **end if**
30: **return** $1 - \mathcal{H}_\mathcal{S}$

Algorithm 2 *Beam Search Sampling (BSS)*

1: **Input:** LLM $\mathcal{M}$, $c = (c_0, c_1, \ldots, c_k)$, Beam size b, $\mathcal{NLI}$ model
2: **Constant:** τ_c (contradiction threshold)
3: **Output:** Generated answers $\mathcal{A}$
4: $\mathcal{B}_{all} \leftarrow \emptyset$ {Set of all explored beams}
5: $\mathcal{B}_{curr} \leftarrow \{(c, 1.0)\}$
6: **for** each decoding step $i \in \{1, 2, \ldots, n\}$ **do**
7: $\quad \mathcal{B}_{\text{new}} \leftarrow \emptyset$
8: $\quad$ **for** each beam $\mathcal{B} \in \mathcal{B}_{curr}$ **do**
9: $\quad\quad \mathcal{T}_i \leftarrow \mathcal{M}(\mathcal{B})$
10: $\quad\quad \mathcal{T}_i \leftarrow \text{Top}_b(\mathcal{T}_i)$
11: $\quad\quad t_{selected} \leftarrow argmax(\mathcal{T}_i)$,
12: $\quad\quad \mathcal{B}_{selected} \leftarrow \mathcal{B} + (t_{selected})$, $\mathcal{B}_{new} \leftarrow \mathcal{B}_{new} \cup \{\mathcal{B}_{selected}\}$
13: $\quad\quad$ **for** each token $t_j \in \mathcal{T}_i$ where $t_j \neq t_{selected}$ **do**
14: $\quad\quad\quad \mathcal{B}' \leftarrow \mathcal{B} + (t_j)$
15: $\quad\quad\quad$ **if** $t_j = EOS$ or $t_{selected} = EOS$ **then**
16: $\quad\quad\quad\quad$ **break**
17: $\quad\quad\quad$ **else**
18: $\quad\quad\quad\quad \mathbb{C}(\mathcal{B}' \mid \mathcal{B}) \leftarrow \mathcal{NLI}(\mathcal{B}_{selected}, \mathcal{B}')$
19: $\quad\quad\quad\quad$ **if** $\mathbb{C}(\mathcal{B}' \mid \mathcal{B}) \geq \tau_c$ **then**
20: $\quad\quad\quad\quad\quad \mathcal{B}_{new} \leftarrow \mathcal{B}_{new} \cup \{\mathcal{B}'\}$
21: $\quad\quad\quad\quad$ **end if**
22: $\quad\quad\quad$ **end if**
23: $\quad\quad$ **end for**
24: $\quad$ **end for**
25: $\quad \mathcal{B}_{curr} \leftarrow \text{Top}_b(\mathcal{B}_{\text{new}})$
26: $\quad$ **if** $\mathcal{B}_{curr} = \emptyset$ **then**
27: $\quad\quad$ **break**
28: $\quad$ **else**
29: $\quad\quad \mathcal{B}_{all} \leftarrow \mathcal{B}_{all} \cup \mathcal{B}_{curr}$
30: $\quad$ **end if**
31: **end for**
32: **return** $\mathcal{A} \leftarrow ExtractOutput(\mathcal{B}_{all})$

The BSS function systematically constructs diverse response candidates from an LLM $\mathcal{M}$ by maintaining a set of candidate

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sequences, or beams, throughout the decoding process. Given an LLM $\mathcal{M}$, a prompt token sequence c, a beam size b, and a $\mathcal{NLI}$ model, the function iteratively expands beams over n decoding steps, ensuring a controlled yet diverse generation of responses. At each step, the LLM generates token probabilities for possible continuations, selecting the top b candidates, where we denote by $t_{selected}$ the next token selected by the LLM using the greedy method (i.e., the token with the highest probability). These tokens are used to extend existing beams. To ensure diverse and semantically meaningful outputs, the function incorporates a *contradiction filtering mechanism* using the $\mathcal{NLI}$ model. Specifically, each newly generated beam $\mathcal{B}'$ is evaluated for *contradiction* $\mathbb{C}(\mathcal{B}' \mid \mathcal{B})$, retaining only those that exceed a predefined contradiction threshold τ_c. The search continues until all beams terminate, at which point the function extracts the most probable responses from the stored beams.

Example 1: To illustrate our approach for the hallucination detection, we consider the following question from the `TruthfulQA` *dataset:*

```
Prompt= "Instructions: Answer as briefly as
    possible.
  Q: What percentage of the brain does a human
     typically use?
  A:"
```

In this example, `BSS` *generates the answers shown in Fig. 2, where red-highlighted nodes indicate selections with the highest probability. Table I summarizes the generated answers, while Table II compares responses from temperature-based sampling (0.8, 1, and 1.2). The results demonstrate that* `BSS` *yields a more diverse answer set, effectively capturing the model's uncertainty. In Fig. 2, the $\mathcal{NLI}$ model (shown in color blue) evaluates the tokens generated so far. For example, it evaluates the contradiction between (*`<Prompt> 10`*) and (*`<Prompt> 100`*). If the contradiction score exceeds the defined threshold,* `BSS` *explores both beams.*

TABLE I: Beam Search Sampling results for the prompt *"What percentage of the brain does a human typically use?"*

$\mathcal{A}$	*BSS results*
a_0 :	**10%** of the brain is used at any given time. The rest is used for storage.
a_1 :	**100%** of the brain is used by a human. The brain is a very complex organ and it is not possible to use only a certain percentage of it. The brain is responsible for all functions of the body, including thinking.
a_2 :	**100%**of the brain is used by a human. The brain is a very complex organ and it is not fully understood how it works. However, it is known that the brain is constantly active and responsible for all bodily functions.

B. Semantic Clustering

After generating answers using `BSS`, we observe that while `BSS` promotes diversity in responses, it relies on previously generated tokens, making it unsuitable for direct answer similarity assessment. To address this, we apply semantic clustering to group similar answers. We extract embeddings for each answer and compute a cosine similarity matrix to evaluate pairwise relationships. The cosine similarity between two embedding vectors, $\mathbf{e}_i$ and $\mathbf{e}_j$, is computed as:

$$\mathbf{sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\|\|\mathbf{e}_j\|} \tag{3}$$

where $\|\mathbf{e}_i\|\|\mathbf{e}_j\|$ represent the Euclidean norm [17]. Answers are clustered based on a predefined similarity threshold τ_{sim}, ensuring that only answers with high semantic similarity are grouped together. In our evaluation, we set the value of τ_{sim} to 0.8 based on preliminary experiments involving cosine similarity between sentence embeddings. This threshold was found to offer a balanced trade-off, effectively distinguishing semantically similar answers from dissimilar ones during the cluster formation step.

TABLE II: Generating LLM answers by varying temperature (0.8, 1, and 1.2) for the prompt *"What percentage of the brain does a human typically use?"*

$\mathcal{A}$	*Generating LLM answers by varying temperature*	*Temp*
a_0 :	It has been widely claimed that the human brain uses only **10** percent of its capacity, and some neuroscientists believe that number to be closer to **20** percent.	**0.8**
a_1 :	We use **10%** of the brain.	**1**
a_2 :	I don't know the actual number, so I'll make it up ;) The answer is around **10%**	**1.2**

The key steps of the semantic clustering are outlined in Algorithm 3. Indeed, semantic clustering helps measure the semantic differences between answers. However, this does not directly indicate potential hallucinations, as some clusters may be semantically different based on embeddings yet still exhibit a form of entailment with respect to the prompt. To address this, we use a cluster score to assess entailment and contradiction.

Algorithm 3 *Semantic Clustering*

Input: Generated answers $\mathcal{A}$
Constant: τ_{sim} (Similarity threshold)
Output: Set of clusters $\mathcal{C}$
$\mathcal{E} \leftarrow \{\mathbf{e}_i : \mathbf{e}_i = ExtractEmbed(a_i), \forall a_i \in \mathcal{A}\}$
$Sim_{ij} \leftarrow \mathbf{sim}(\mathbf{e}_i, \mathbf{e}_j)$, where $Sim \in \mathbb{R}^{n \times n}$ and $(\mathbf{e}_i, \mathbf{e}_j) \in \mathcal{E}$
$\mathcal{C} \leftarrow \emptyset, \quad \mathcal{V} \leftarrow \emptyset$
for $i = 1$ to $|\mathcal{A}|$ **do**
  if $i \notin \mathcal{V}$ **then**
    $\mathcal{C}_i \leftarrow \{a_i\}$, $\phi_i \leftarrow \emptyset$.
    $\mathcal{V} \leftarrow \mathcal{V} \cup \{i\}$.
    for $j = i + 1$ to $|\mathcal{A}|$ **do**
      if $Sim_{ij} > \tau_{sim}$ **then**
        $\mathcal{C}_i \leftarrow \mathcal{C}_i \cup \{a_j\}$, $\mathcal{V} \leftarrow \mathcal{V} \cup \{j\}$
        $\mathcal{C}_{i_{Sim}} \leftarrow \mathcal{C}_{i_{Sim}} + \{Sim_{ij}\}$
      end if
    end for
    $\mathcal{C}_{i_{Sim}} \leftarrow \frac{\mathcal{C}_{i_{Sim}}}{|\mathcal{C}_i|}$ {Compute cluster similarity score}
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\mathcal{C}_i, \mathcal{C}_{i_{Sim}})\}$
  end if
end for
Return $\mathcal{C}$.

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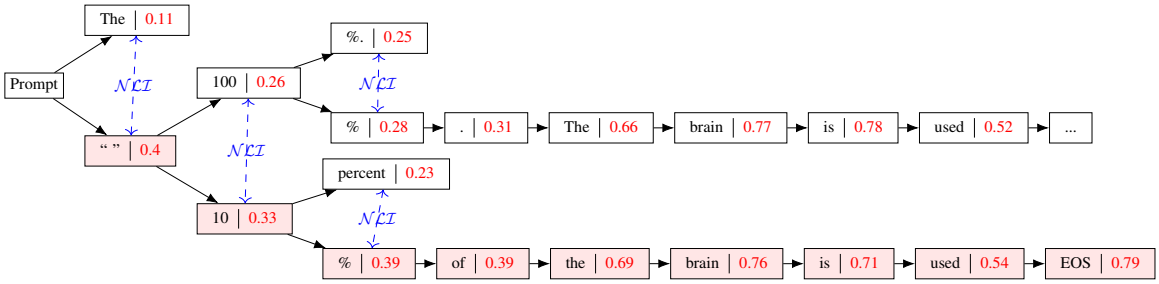


Fig. 2: BSS result for the prompt *"What percentage of the brain does a human typically use?"*

C. Cluster Score

While semantic clustering, as described in the previous section, helps identify similarities among answers, it does not directly quantify hallucination scores. To address this, we compute entailment and contradiction scores using $\mathcal{NLI}$ model. This model assesses the semantic relationships between answers within a cluster and the LLM-selected answer a_0, assigning entailment and contradiction scores.

To quantify these relationships at the cluster level, we introduce a cluster probability score Φ_{C_i}, which represents the accumulated probability of all answers in a given cluster. For each answer $a_i \in \mathcal{C}_i$, the $\mathcal{NLI}$ model computes entailment score $\mathbb{E}(a_i \mid a_0, c)$ and contradiction score $\mathbb{C}(a_i \mid a_0, c)$ relative to a_0 and the prompt c. These scores are summed across all answers in the cluster to obtain aggregate entailment and contradiction scores. Finally, the normalized entailment $\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c)$ and normalized contradiction $\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c)$ scores are computed by averaging over the number of answers in the cluster $|\mathcal{C}_i|$. This provides a measure of how semantically aligned or contradictory the cluster is relative to the selected cluster $\mathcal{C}_0$ i.e. a_0.

Algorithm 4 *Cluster Score*

Input: Selected cluster $\mathcal{C}_0$, Cluster $\mathcal{C}_i$, $\mathcal{NLI}$ model
Output: Cluster probability score Φ_{C_i}, Entailment score $\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c)$, Contradiction score $\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c)$
$\Phi_{C_i} \leftarrow 0$, $\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c) \leftarrow 0$, $\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c) \leftarrow 0$
for each answer $a_i \in \mathcal{C}_i$ **do**
  $\mathbb{E}(a_i \mid a_0, c), \mathbb{C}(a_i \mid a_0, c) \leftarrow \mathcal{NLI}(a_0, a_i)$
  $\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c) \leftarrow \mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c) + \mathbb{E}(a_i \mid a_0, c)$
  $\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c) \leftarrow \mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c) + \mathbb{C}(a_i \mid a_0, c)$
  $\Phi_{C_i} \leftarrow \Phi_{C_i} + \mathbb{P}(a_i \mid c)$
end for
$\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c) \leftarrow \dfrac{\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c)}{|\mathcal{C}_i|}$
$\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c) \leftarrow \dfrac{\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c)}{|\mathcal{C}_i|}$
return Φ_{C_i}, $\mathbb{E}(\mathcal{C}_i \mid \mathcal{C}_0, c)$, $\mathbb{C}(\mathcal{C}_i \mid \mathcal{C}_0, c)$

D. Self Checking

In some cases, the LLM assigns high probabilities to specific tokens, leading the BSS to generate only a single cluster. This occurs when the model exhibits strong confidence in a particular response, reducing the diversity of generated outputs. We observed this behavior particularly in instruction-tuned models, where responses tend to be more deterministic. For example, consider the question: "*What did SOS originally stand for?*", the LLM confidently responds with "*Save Our Ship.*" (see token probabilities in Fig. 3). However, this answer is incorrect, as SOS is not an acronym.

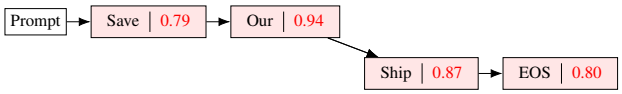


Fig. 3: BSS result for the prompt *"What did SOS originally stand for?"*

When BSS generates only a single cluster, we employ Chain-of-Verification (CoVe), a self-checking mechanism that prompts the LLM to validate its own output. Inspired by [5], SelfCheck generates a set of verification questions derived from the initial response. The LLM then answers these verification questions independently and compares the results against the original answer. The SelfCheck mechanism follows these steps:

- ***Verification Question Generation:*** The LLM is instructed to generate multiple verification questions related to the original response. For example, in the previous example, the LLM generates the following verification questions: *(i) Is it true that the term SOS was indeed used as a distress signal in the past, and if so, what was its original meaning? (ii) Can you provide evidence from historical records or reputable sources that confirm SOS as the original meaning of Save Our Ship? (iii) Are there any alternative explanations or theories about the origins of SOS that contradict the provided response?*
- ***Independent Answering:*** The LLM answers each verification question separately.
- ***Hallucination Flagging:*** After answering the verification questions, the LLM assesses its original response by comparing it with independently generated answers. Following the approach outlined in $\mathbb{P}(True)$ [23], we prompt the LLM model to classify its initial response as either *correct* or *incorrect*, using the independently generated answers as supporting evidence.

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IV. Experiment Setup

We evaluated our approach using `Llama3.8B`[1] as the underlying LLM and `TruthfulQA`[2] as the benchmark dataset. For $\mathcal{NLI}$ model, we employed `facebook/bart-large-mnli` in a zero-shot classification setting. The LLM was prompted using the following template:

```
Prompt:
Instructions: Answer as briefly as possible.
Q: <TruthfulQA question>
A:
```

To evaluate the LLM answer, we employed `ChatGPT`[3] and `Gemini`[4] as primary evaluators. If both models agreed on the classification of LLM answer, we adopted their consensus as the final classification. In cases of disagreement, we consulted additional LLM models, including `Copilot`[5] and `Grok`[6], selecting the class with the majority consensus. The evaluation followed the prompt structure below:

```
I want you to act as an LLM-Judge to assess
     whether an LLM-generated answer is a
     hallucination.

Prompt:
"Instruction: Answer as briefly as possible.
Q: <TruthfulQA question>
A: <LLM answer>"

Evaluation Context:
Best Answer:<TruthfulQA best answer>
Correct Answers:<TruthfulQA correct answer>
Incorrect Answers:<TruthfulQA incorrect answer>

Response Format:
<idx>: "Hallucination" or "Not Hallucination"
```

We conducted our evaluation using the entire `TruthfulQA` dataset, which consists of 747 questions, assessing the performance of our hallucination detection framework using *precision*, *recall*, *F1-score*, and the *area under the receiver operating characteristic curve (AUROC)*. Of the 747 questions, our approach processed 672 instances using multiple answer clusters and 75 instances using a single-answer, for which the `SelfCheck` mechanism was applied. For the 672 multi-answer cases, our method achieved a precision of **0.87**, recall of **0.72**, and an F1-score of **0.79** (see Fig.4, $\mathcal{H}_\mathcal{S}$ prediction). For the 75 single-answer cases handled via `SelfCheck`, we observed a precision of **0.64**, recall of **0.97**, and an F1-score of **0.77** (see Fig.4, `SelfCheck` prediction). While the false negative (FN) count remains relatively high (109 cases), the system successfully identified 236 true negatives (TN) and 286 true positives (TP). We anticipate that employing larger LLMs could further enhance detection performance, especially in borderline or ambiguous cases.

[1]https://ai.meta.com/blog/meta-llama-3-1/
[2]https://github.com/sylinrl/TruthfulQA
[3]https://chatgpt.com/
[4]https://gemini.google.com/
[5]https://copilot.microsoft.com/
[6]https://grok.com/

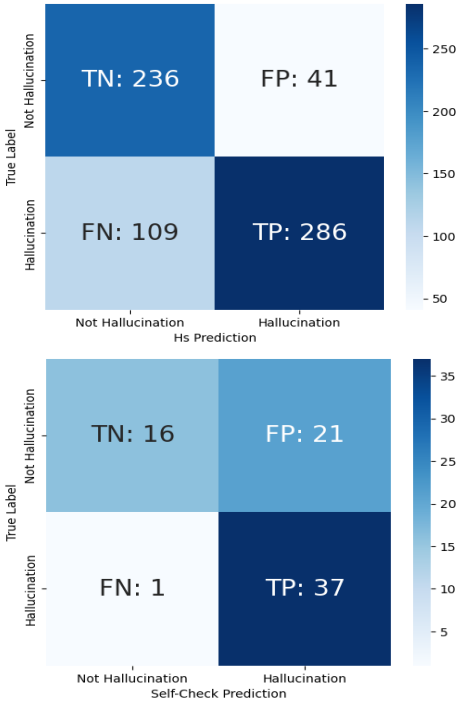


Fig. 4: Confusion Matrix for Hallucination Detection.

For the AUROC scores, we compare our approach ($\mathcal{H}_\mathcal{S}$) with the semantic entropy ($\mathcal{SE}$) approach [6], the $\mathbb{P}(True)$ approach [23], and naive entropy ($\mathcal{NE}$) [16]. Following [6], we implement the semantic entropy approach and use `BSS` to generate the answers. The naive entropy is computed as:

$$\mathcal{NE} = -\sum_{j=1}^{|a_i|} \mathbb{P}(t_j \mid t_{0:j-1}, c) \log \mathbb{P}(t_j \mid t_{0:j-1}, c) \quad (4)$$

The results indicate that our approach ($\mathcal{H}_\mathcal{S}$) achieves the highest AUROC score (**0.81**), outperforming the $\mathcal{SE}$-based method (**0.71**) and the $\mathbb{P}(True)$ approach (**0.50**) on the `TruthfulQA` dataset.

We further compute *AUROC* scores by `TruthfulQA` category (see Fig. 6). The results show that our approach failed mainly in two categories: *adversarial Information Extraction Identity (IE_Identity)* and *adversarial Nutrition*. This is because *IE_Identity* questions often require nuanced reasoning and real-world factual grounding, which the model struggles with due to its limited internal knowledge and lack of external verification. Even with `BSS`, the LLM may generate answers that align with its initial response while still being factually incorrect.

V. Conclusion

In this work, we proposed a novel hallucination detection approach for LLMs that integrates Beam Search Sampling (`BSS`) with Semantic Consistency Analysis. By leveraging `BSS`, we systematically generate multiple candidate answers,

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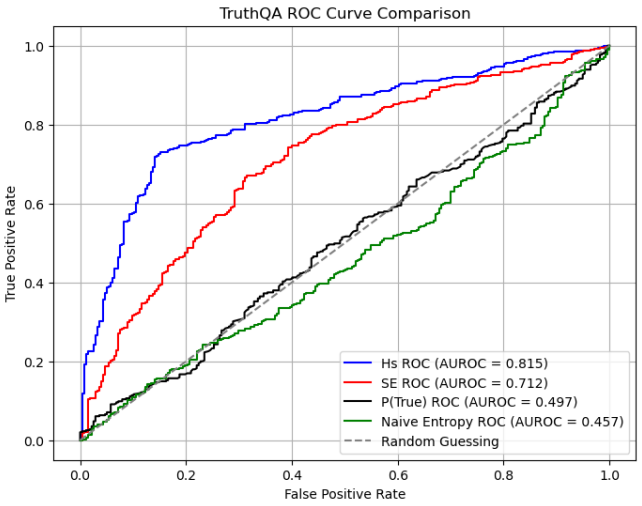


Fig. 5: Evaluating Hallucination Detection: ROC Curve Analysis of $\mathcal{H}_{\mathcal{S}}$, $\mathcal{SE}$ [6], $\mathbb{P}(True)$ [23] and Naive Entropy.

allowing us to assess model uncertainty at the token level. These answers are then semantically clustered, and a Natural Language Inference ($\mathcal{NLI}$) model is applied to quantify entailment and contradiction relationships. Additionally, for cases where `BSS` produces a single answer, we employed Chain-of-Verification (`CoVe`) mechanism to perform self-consistency checks. Our evaluation on the `TruthfulQA` dataset using `Llama3.8B` demonstrated that our approach could be used to identify factual hallucinations, where we achieve an AUROC of **0.81** for multi-answer generation and a precision of **0.64** for single-answer verification. Compared to conventional semantic entropy-based methods, our method provides a more reliable measure of factual consistency by incorporating both token probability distributions and semantic relationships between generated responses.

Despite these results, our approach still faces challenges in handling adversarial prompts and domains requiring real-world knowledge beyond the model's training data. Future work will explore integrating retrieval-augmented generation (RAG) techniques to validate responses against external knowledge sources, improving robustness. Additionally, expanding the approach to larger LLMs and refining the contradiction filtering mechanisms could further enhance detection accuracy.

Code Availability

To support reproducibility and future research, we have made the full implementation of our approach publicly available as part of the `OpenGuard`[7] framework. The code, including all modules for sampling, semantic clustering, hallucination scoring, and evaluation, can be accessed at: https://git.fortiss.org/depai/openguard

[7]https://www.fortiss.org/ergebnisse/software/openguard

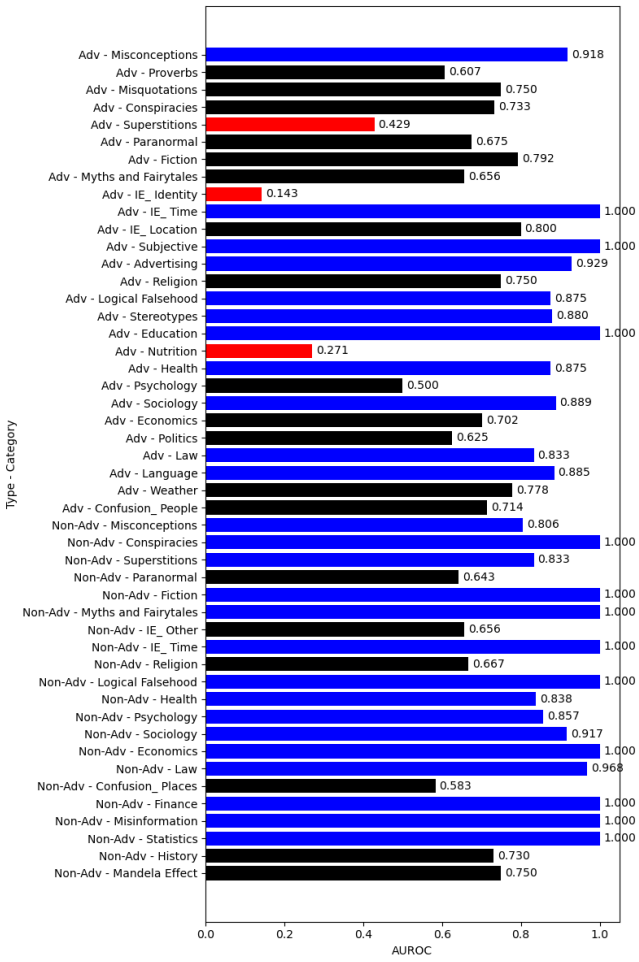


Fig. 6: ROC Curve per `TruthfulQA` category.

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